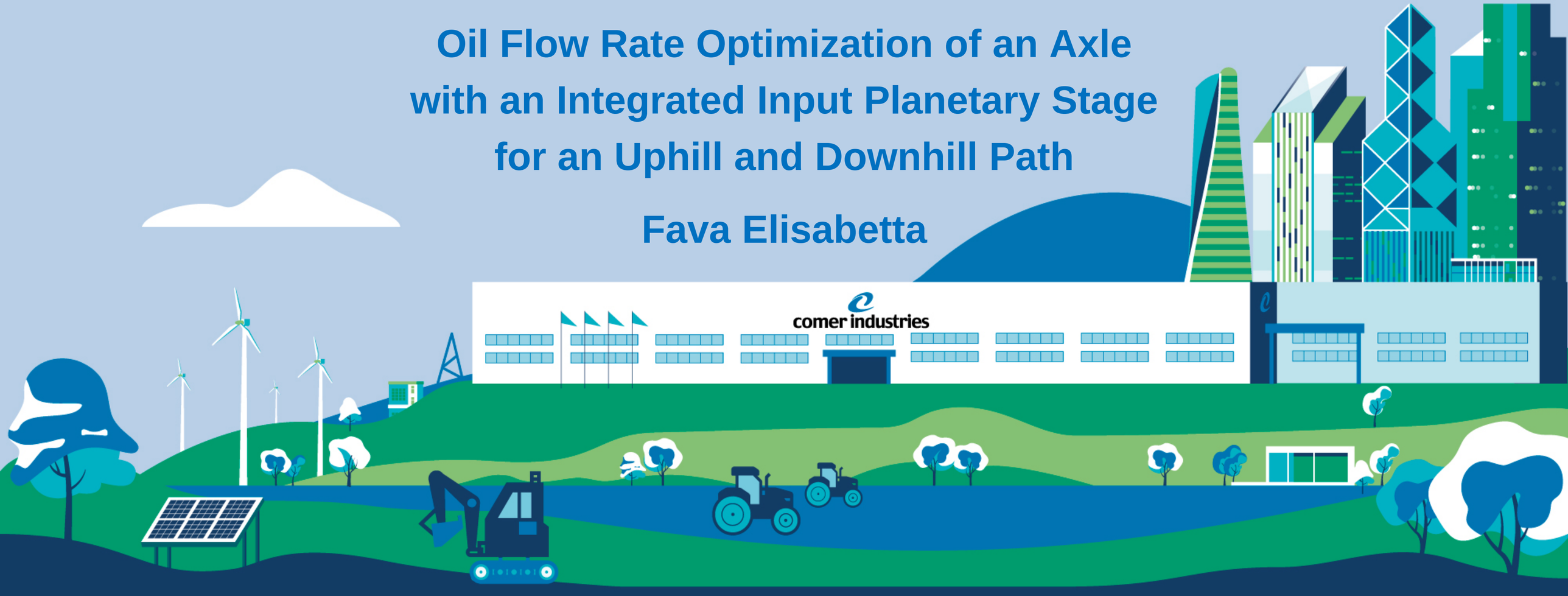
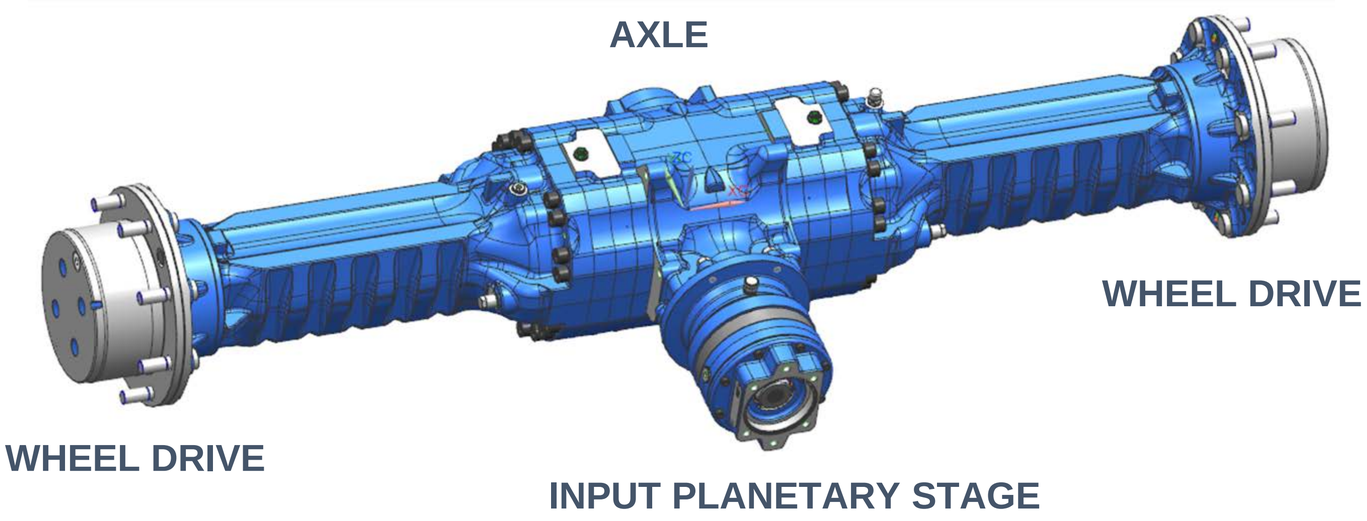
comer industries

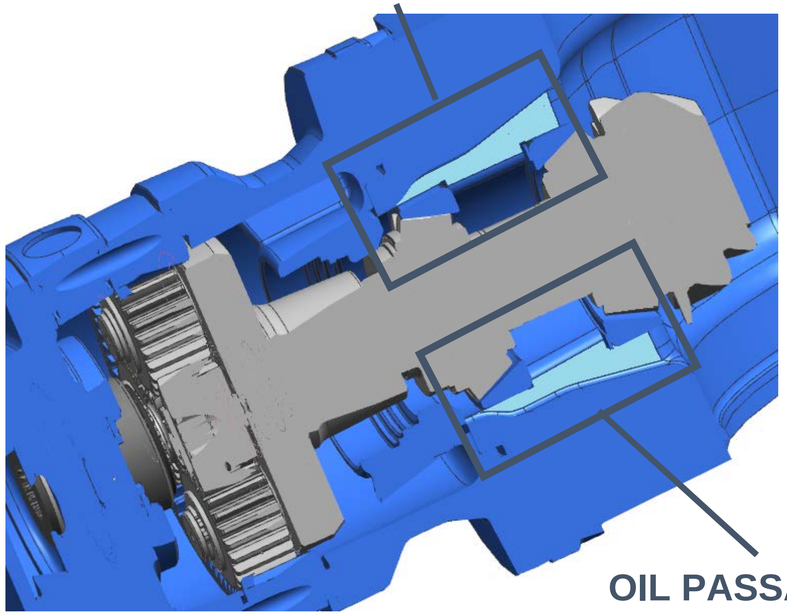
Oil Flow Rate Optimization of an Axle with an Integrated Input Planetary Stage for an Uphill and Downhill Path

Fava Elisabetta



Case Study





INPUT PLANETARY STAGE

Application

Soil Compactor

Targets

Optimization of the oil passages geometry to guarantee:

- the **proper lubrication** of the input planetary stage
- the **proper oil exchange** between the input planetary stage and the axle able to avoid a rise of temperature in the planetary stage without worsening the axle efficiency

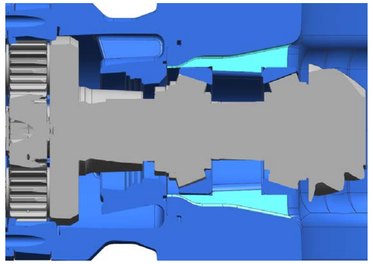
when the machine goes **uphill** and **downhill**

Case Study

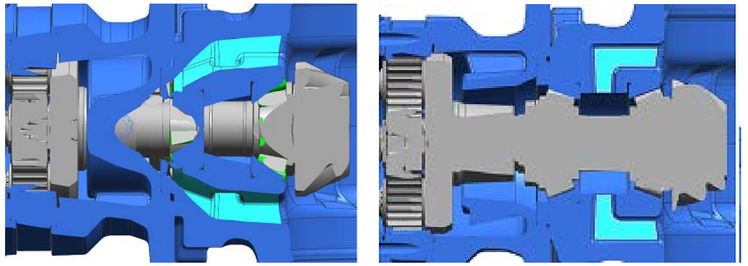
Case	Name	Oil passage geometry	Oil level
1	Baseline	Current	Current
2	Opt	Optimized	Current
3	Opt_High	Optimized	Increased

OIL PASSAGE GEOMETRY

CURRENT



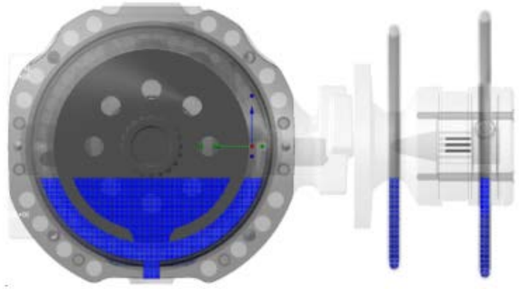
OPTIMIZED



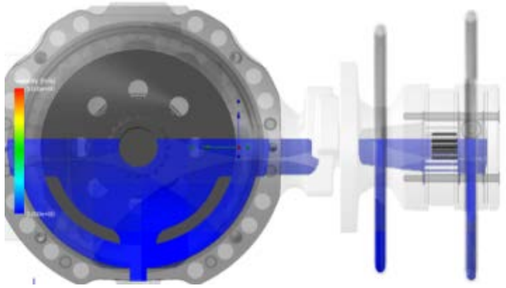
Two different sections of the side passages

- The current oil passage geometry and an optimized geometry set with the current oil level have been compared
- The optimizations consist of a reduction of the side passage section between the bearings and an enlargement of the side passage section to the planetary stage
- A higher oil level has been verified to evaluate how it affects the oil flow

OIL LEVEL

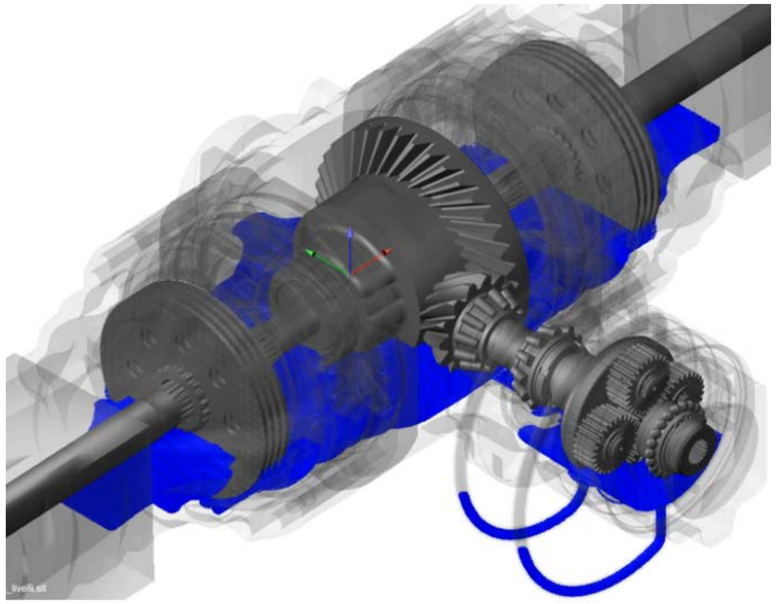


Current: 6,0 l



Increased: 9,5 l

Model Building – Domain and Oil Data

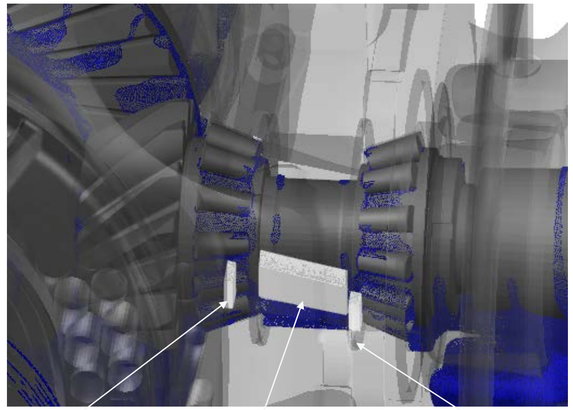


- The housings and all the internal components of the axle and the planetary stage have been modeled
- Oil domain has been reduced closing the internal volume just after the brake disks: the deleted areas are far enough from the spiral gear and the planetary stage
- Two external pipes have been added to evaluate the dynamic oil level
- Oil data:

  ν = 44,2 cSt
  T = 65 °C
  ρ = 950 kg/m3
  γ = 0,03 N/m

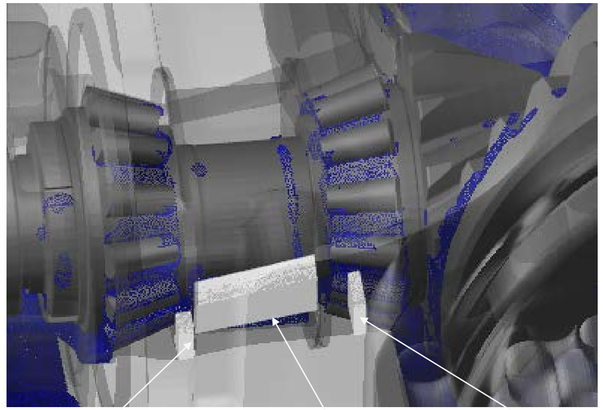
- To evaluate the flow rates in the passages and towards the planetary and between the bearings three probe regions for the right side (crown side) and for the left side have been introduced
- The input flow region shows the oil amount from the axle to the oil passage
- The bearing flow region shows the oil amount that goes in the area between the two bearings
- The planetary flow region shows the oil amount that goes in the planetary

LEFT SIDE



Input Flow | Bearings Flow | Planetary Flow

RIGHT SIDE

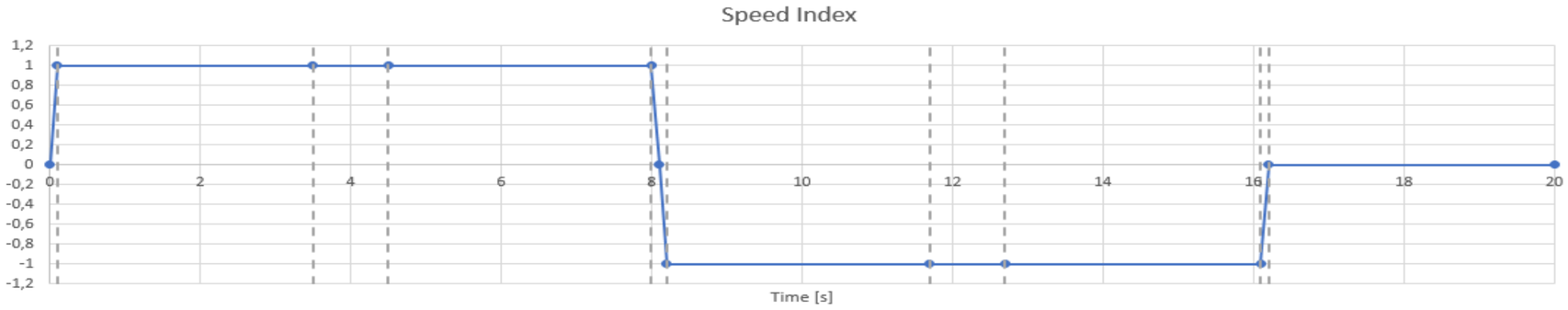


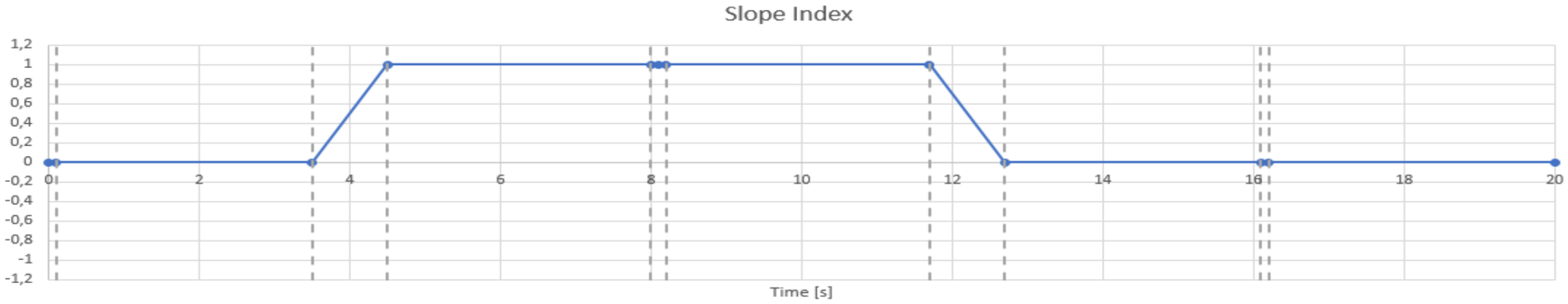
Planetary Flow | Bearings Flow | Input Flow

Model Building – Machine Path

For all the three simulations, the same speed and slope profiles have been defined to reproduce an uphill and downhill path

Time [s]	Angle [°]	Input Speed [rpm]
0,0	0	0
0,1	0	1330
3,5	0	1330
4,5	17	1330
8,0	17	1330
8,1	17	0
8,2	17	-1330
11,7	17	-1330
12,7	0	-1330
16,1	0	-1330
16,2	0	0
20,0	0	0





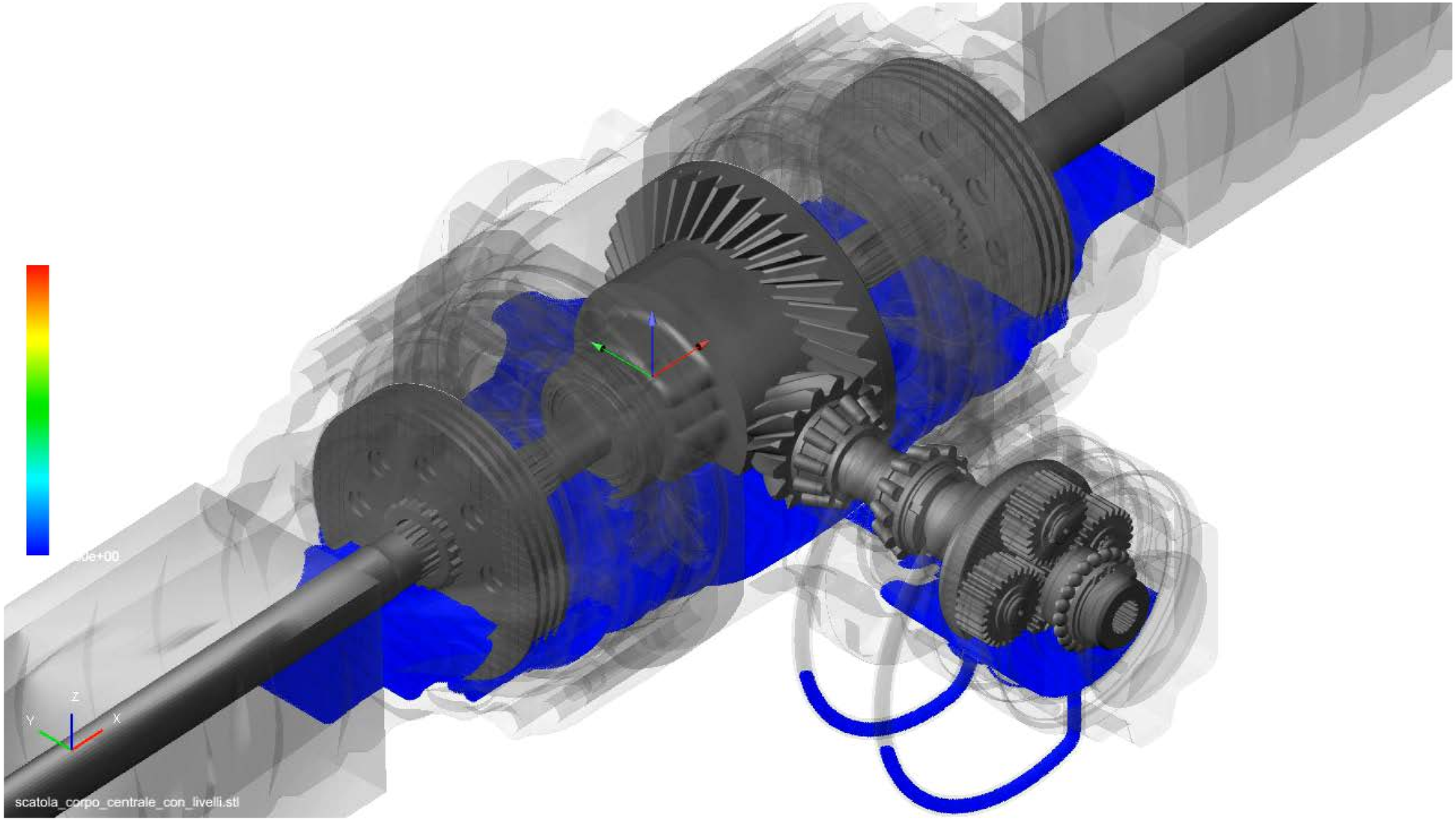
Horizontal Forward	Uphill	Downhill	Horizontal Rearward	Steady

Model Setup

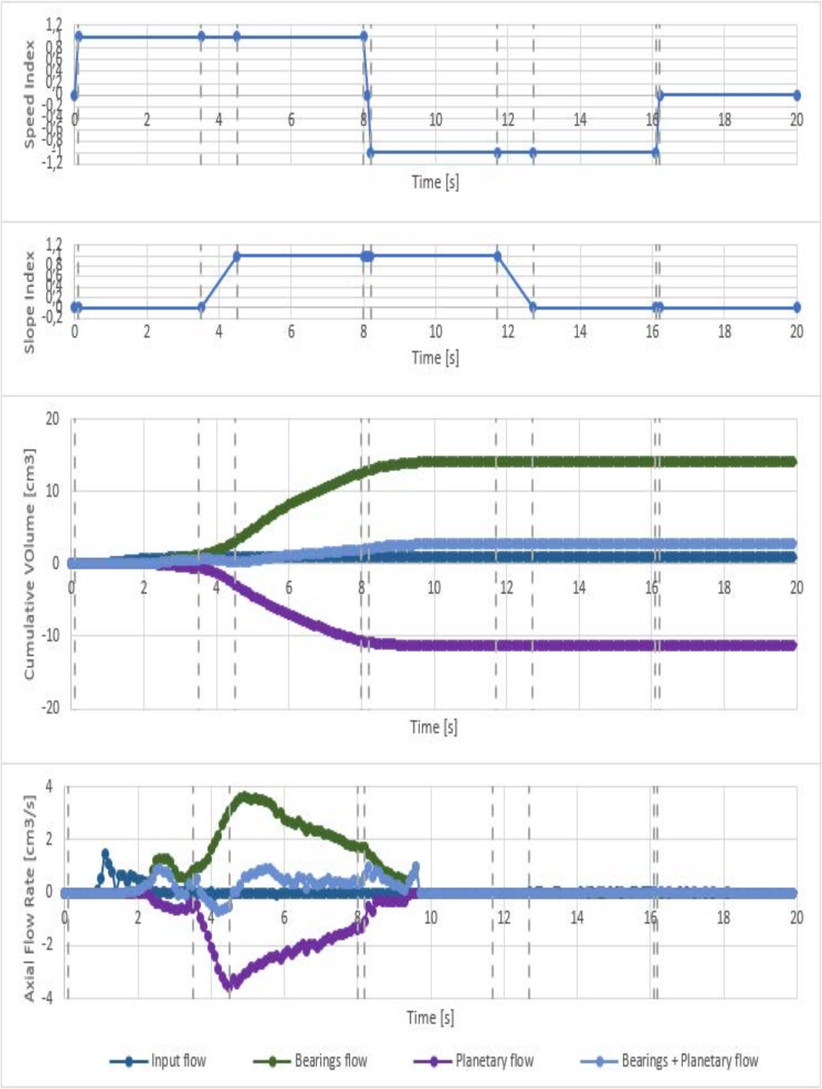
Simulation Parameters	Values
Courant Number	0,20
Speed of Sound [m/s]	10
Turbulence	ON
Slip Factor – Internal Components	5
Slip Factor - Housings	1
Particle Size [mm]	1,00
Pressure Smoothing	OFF
Surface Tension Model	Potential

- 6.000.000 particles for case 1 and case2
  9.500.000 particles for case 3
- Time step : 5,8 E-05 s
- 2 simulation sec / 1 day on average (12 days per case study)
- Running on a NVIDIA K80 Graphic Card 12 GB GRAM

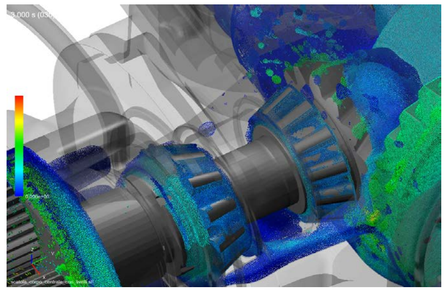
Results – Overall Lubrication



Results – Baseline Case Study



RIGHT SIDE – CROWN SIDE

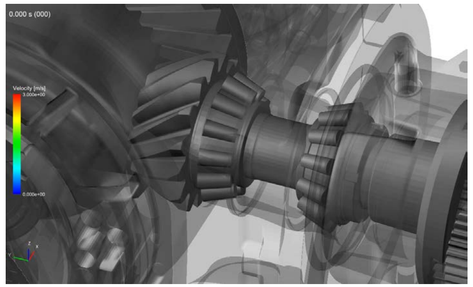


On the right side:

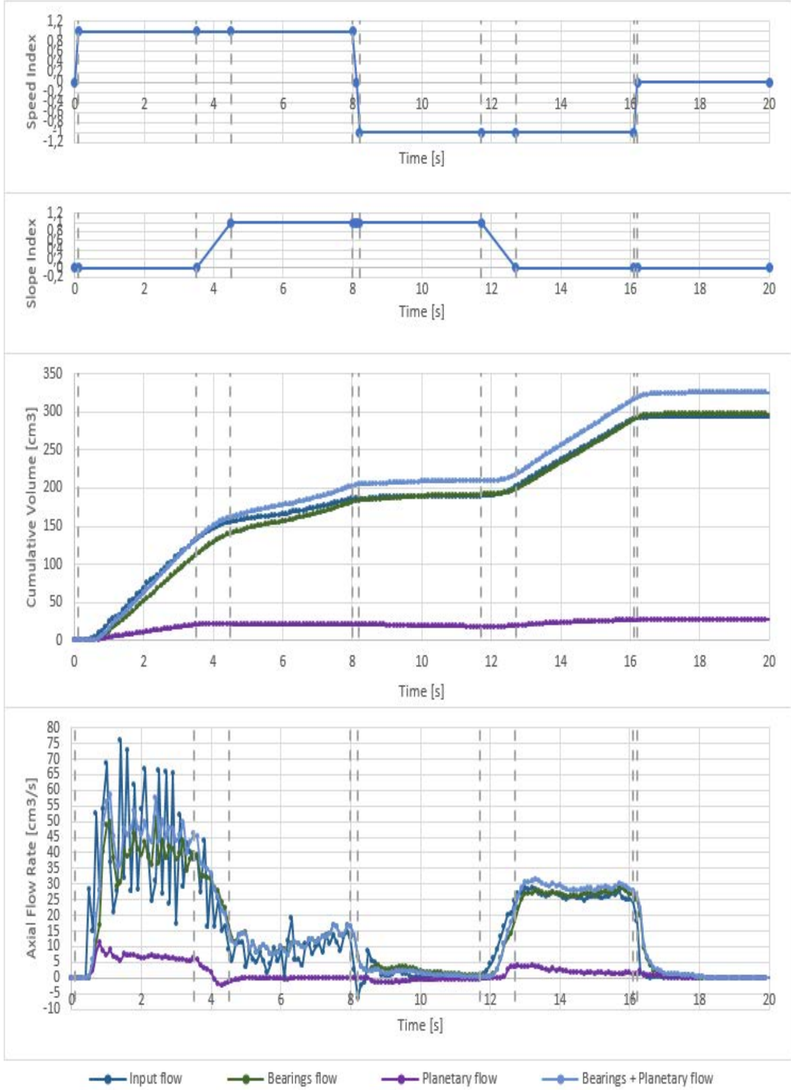
- The crown is able to force the oil into the passage in each configuration (blue curve)
- Most of the oil however goes between the two bearings (green curve) and only a small amount reaches the planetary (violet curve)

On the left side:

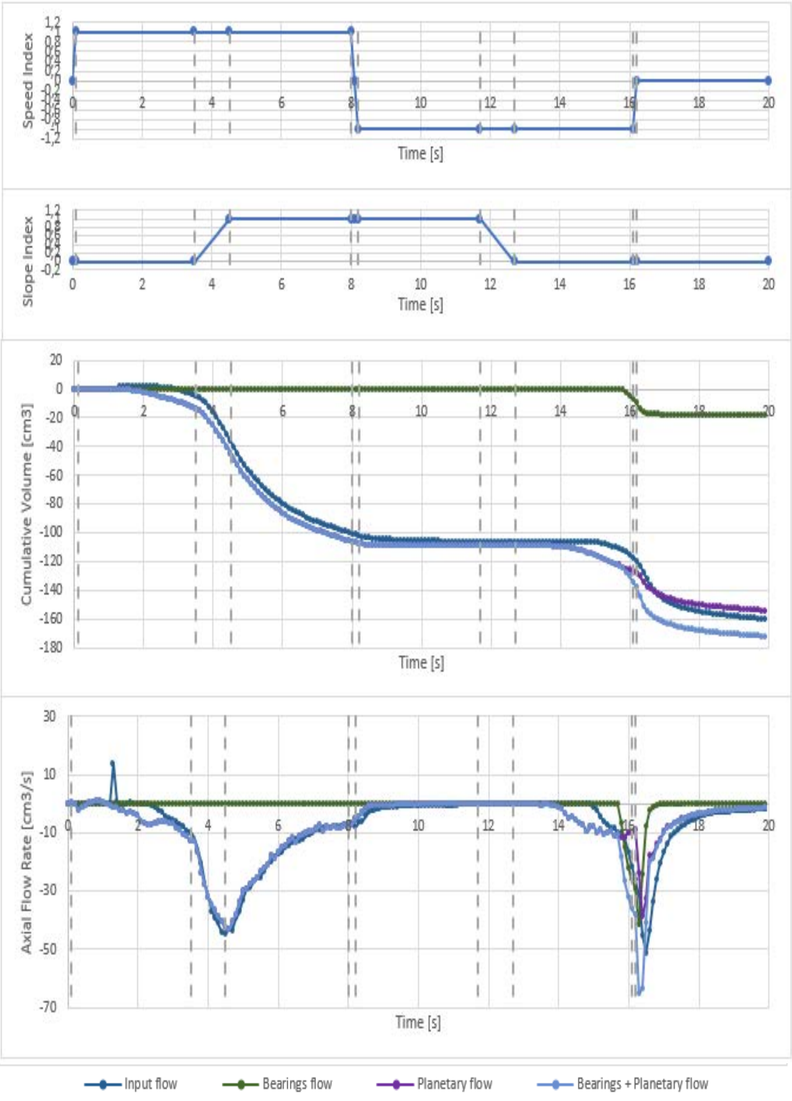
- The input flow is almost null
- When the axle slopes uphill, a small amount of oil comes out of the planetary and goes between the bearings



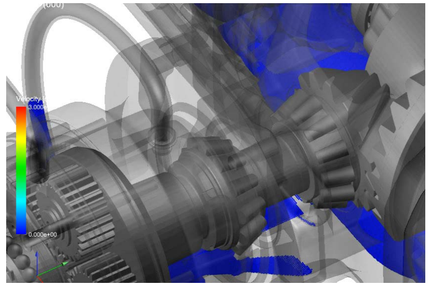
LEFT SIDE



Results – Opt Case Study



RIGHT SIDE – CROWN SIDE

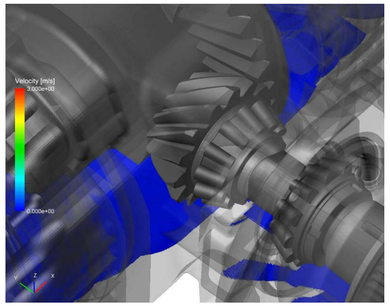


On the right side:

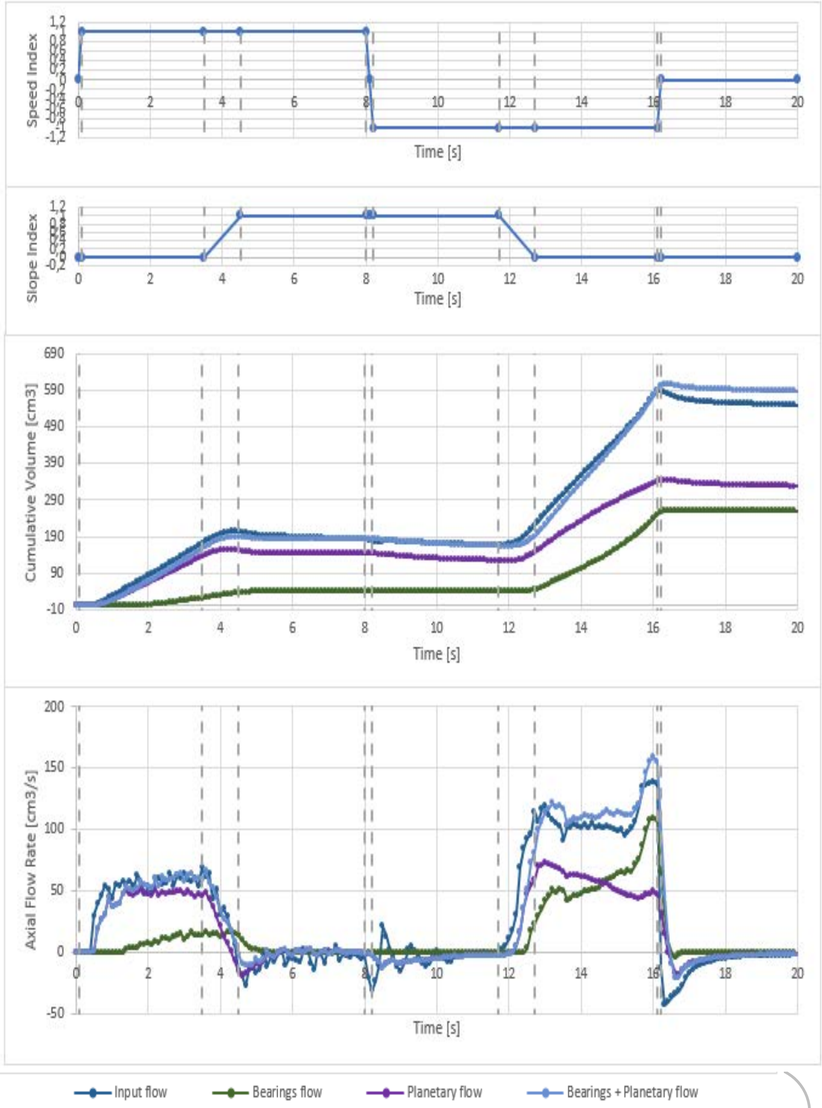
- The input flow rate is increased compared to the baseline case
- The planetary is filled with an increased oil amount, comparable in the horizontal forward and rearward condition
- The area between the two bearings is filled especially in rearward condition
- When the axle is sloped no oil reaches the planetary and the bearings

On the left side:

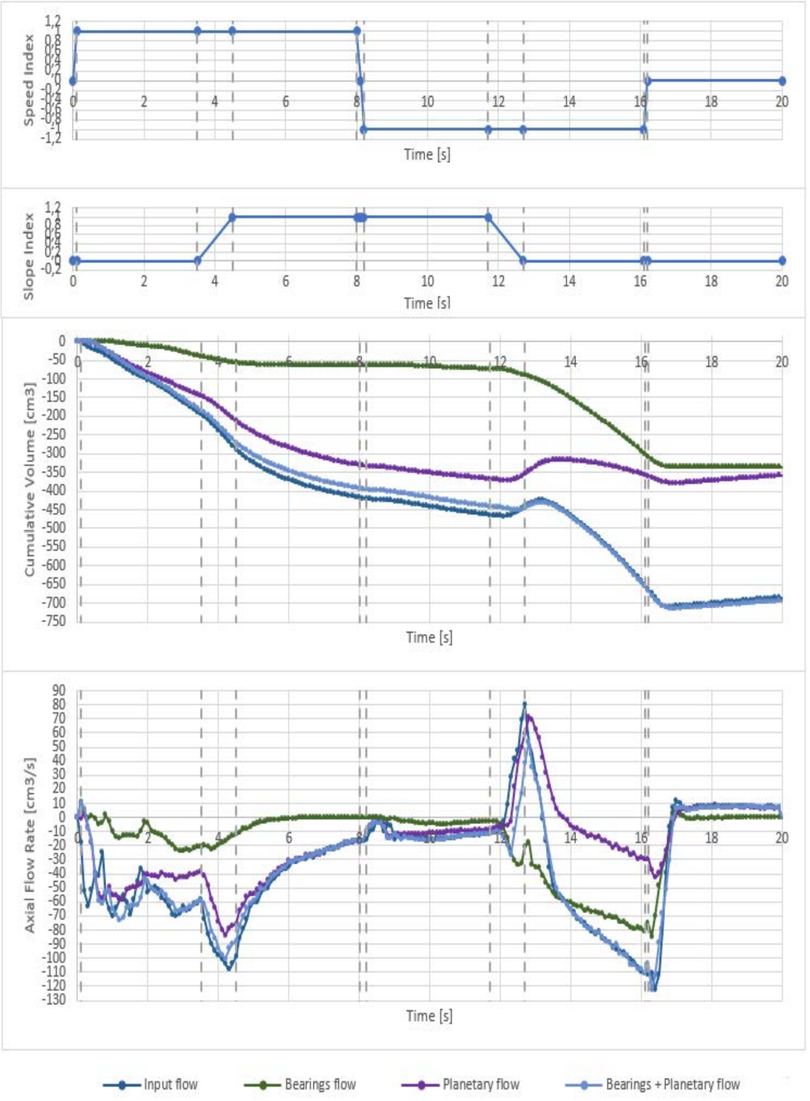
- When the axle is sloping uphill, a small amount of oil comes out of the planetary and goes toward the axle
- When the axle stops, the particles left in the passages and between the rollers goes into the axle



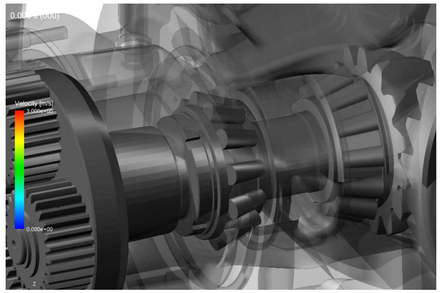
LEFT SIDE



Results – Opt_High Case Study



RIGHT SIDE – CROWN SIDE

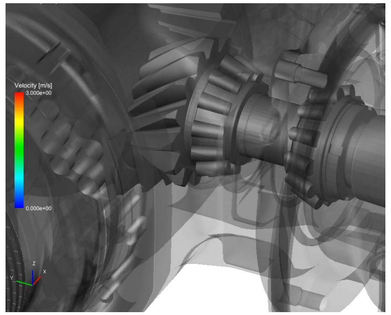


On the right side:

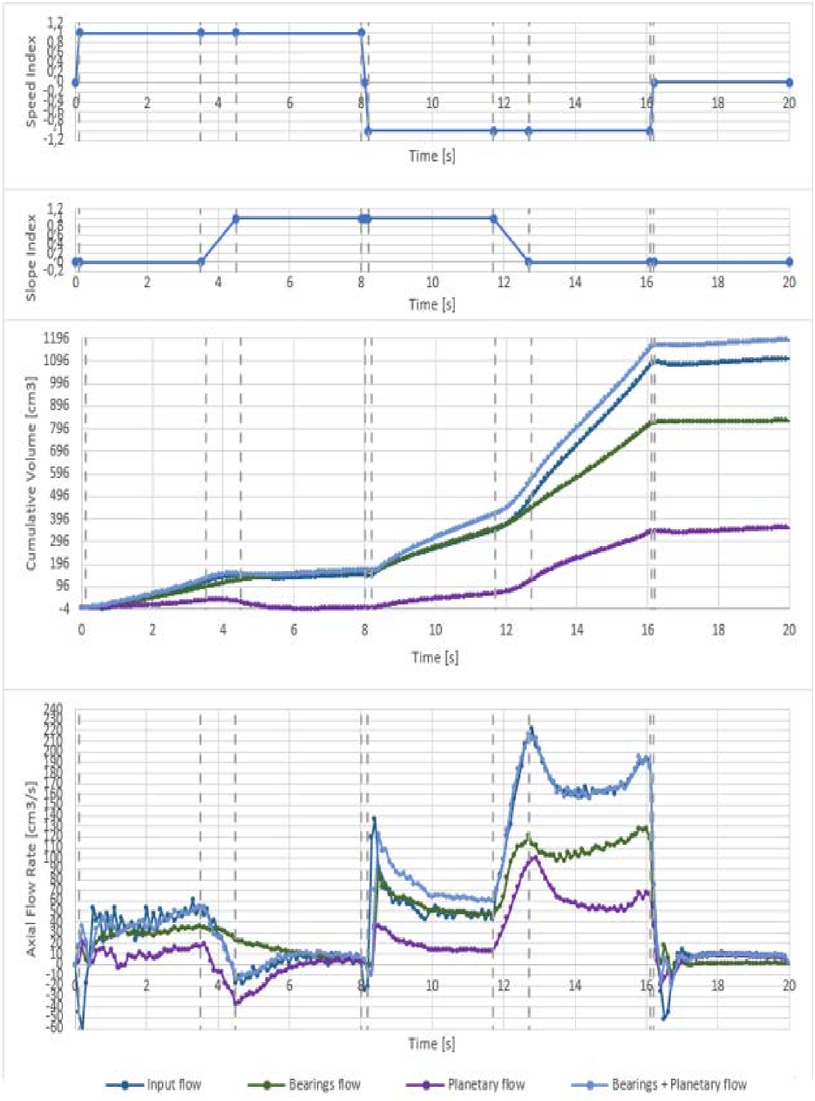
- The input flow rate is increased compared to the previous cases especially in rearward
- The flow rate to the planetary is slightly increased, while is strongly increased the flow rate between the bearings
- When the axle is uphill a small oil amount goes from the planetary to the passage between the bearings

On the left side:

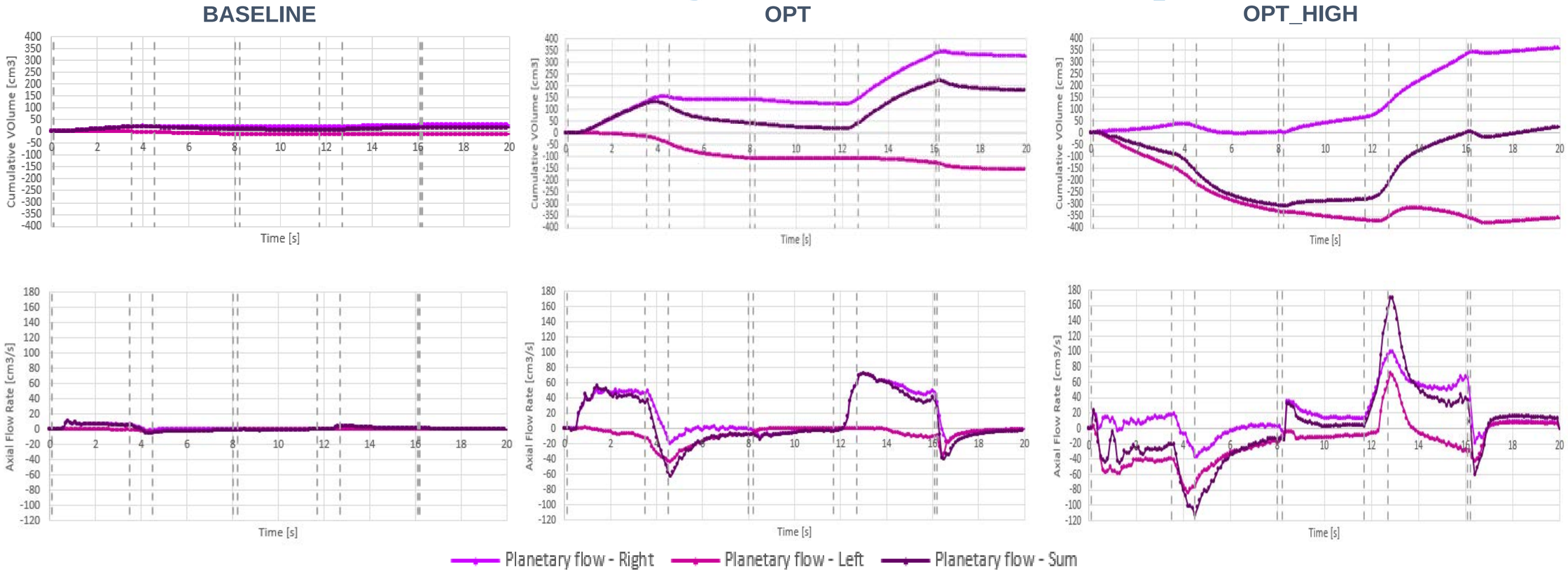
- The oil always comes out from the planetary and the bearings
- The flow rate positive spike when the axle returns horizontal is due to the change of direction of the particles already present in the passage
- The oil comes out of the bearing area especially in horizontal position because it is fulfilled by the right-side passage



LEFT SIDE



Results – Planetary Flow Comparison



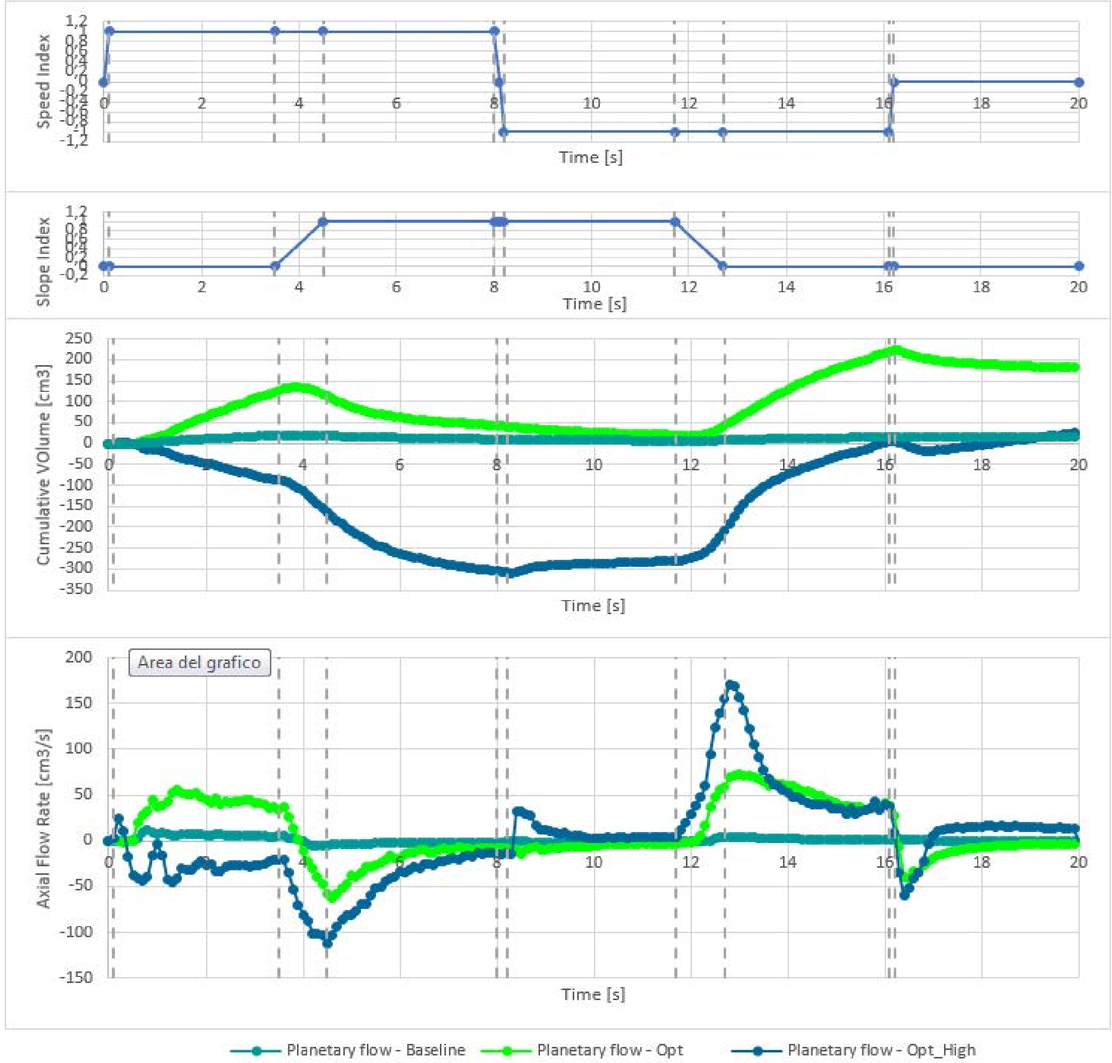
Positive volumes and flows mean that the particles are entering the area

- There is a very small oil exchange in the Baseline case in each phase of the simulation
- With the optimized geometry a greater oil amount reaches the planetary from the right passage (purple curve), but there is also a part that leaves the planetary through the left passage (pink curve)
- For the Opt case, the overall oil quantity in the planetary increases compared to the initial level (dark violet curve)
- For the Opt_High case, the overall oil quantity in the planetary at the end of the simulation is the same compared to the initial level (dark violet curve): initially the oil quantity that goes out of the planetary is greater than the one that comes in, then, when the axle is in the downhill phase and the following horizontal position, the oil returns to the initial value in the planetary.

Results – Planetary Flow Comparison

Comparing the overall oil quantity (sum of right and left oil quantity):

- The Opt case (brilliant green curve) highlights a significant variation of oil volume during the horizontal phases. Since the initial oil level is the same of the baseline case, the geometry modifications give significant improvements
- The Opt_high case (Blue curve) is always negative meaning that the amount of oil is never greater than the initial one and goes out of the planetary for the first part of the simulation. However, this doesn't mean that the oil quantity is lower than the other two cases because the initial level is higher. (The cumulative volume graph shows the oil quantity variation trend regardless of the different initial level)
- The Opt_high case (Blue curve peak to peak) guarantees the greatest oil exchange between the axle and the planetary



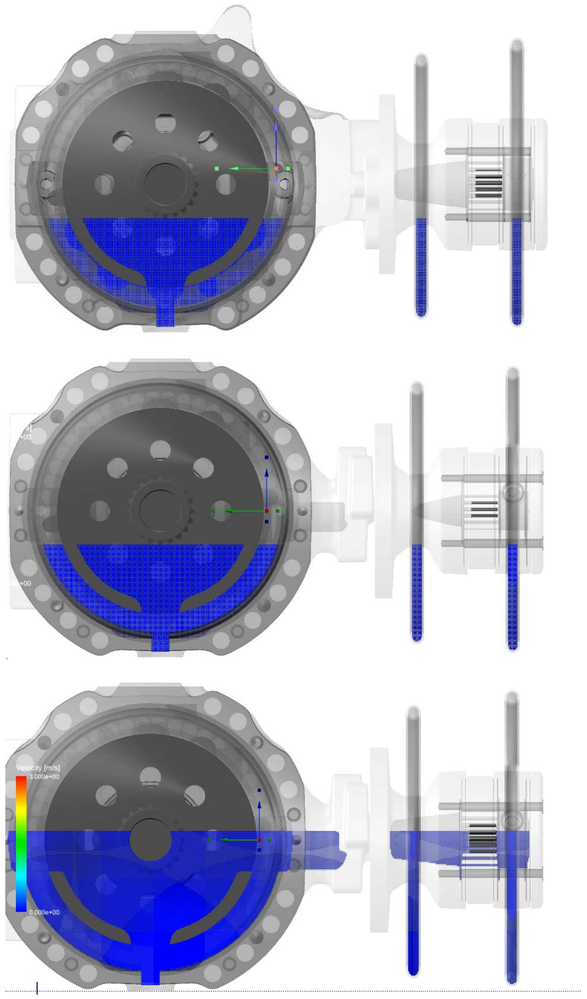
Positive volumes and flows mean that the particles are entering the area

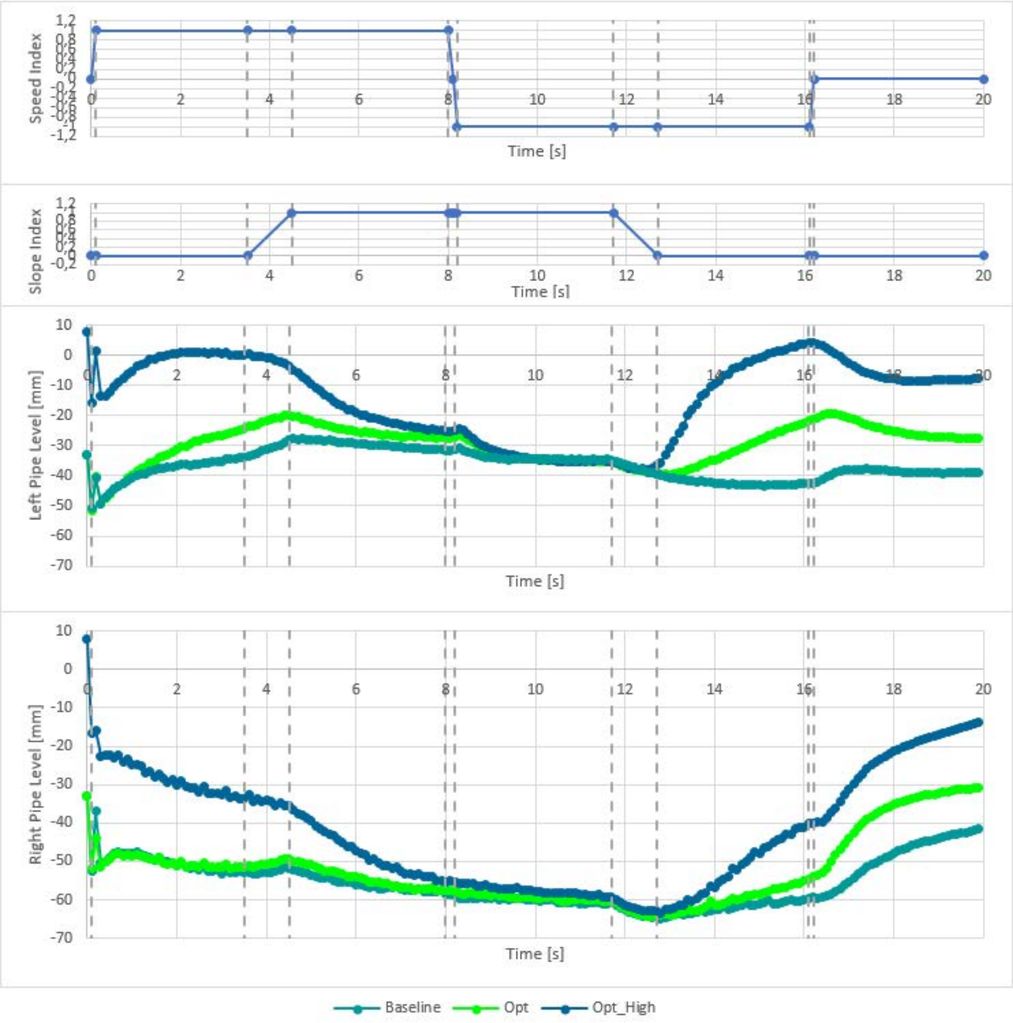
Results – Oil Level Comparison

BASELINE

OPT

OPT_HIGH

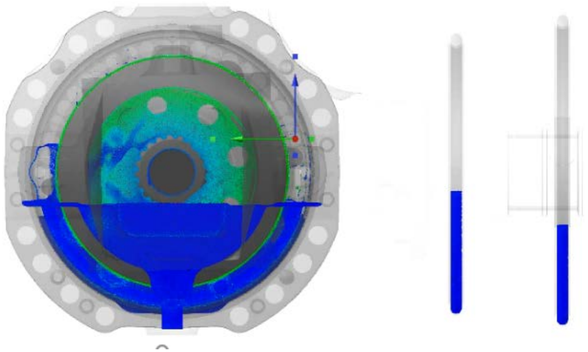


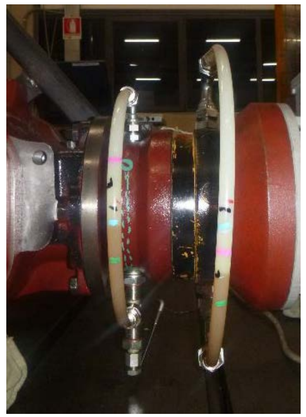


The oil level in the two pipes confirms that the Opt_High case has never a lower oil quantity in the planetary compared to the other two cases and, since the oil level is the same for the three simulations when the axle is sloped, it is also confirmed that the oil exchange is the greatest for this latter case.

Conclusions

- Three simulations has been run reproducing a machine path made by a horizontal forward phase followed by un uphill and a downhill phase and a horizontal rearward one.
- The current oil passages geometry has shown an inadequate oil exchange between the axle and the planetary: the planetary is lubricated but there is no oil recirculation between the two areas. This could give some issues in terms of local temperature in the planetary area.
- Optimizing the oil passages geometry with the same oil level, a significant improvement in the oil exchange has been highlighted especially during the horizontal phases
- Increasing the oil level, the oil exchange is further improved but, with this new oil level, the oil goes out of the planetary for most of the simulation reaching the same level of the other two cases when the axle is sloped: this means that the level is too high and could penalize the efficiency
- The most suitable solution is to maintain the same level of oil and to implement the geometry modifications
- A verification of the simulation results has been done in the test lab mounting two pipes and comparing the oil levels: a very good correspondence has been found for the dynamic level in horizontal position, before and after the sloped steps.





comer industries

Thank you for your attention